

Due by 10:00 PM 04/03. Web uploading.

J.D. Jackson, Chapter 6 (3rd edition)

1. (20) Transformation property of E&M and the Hall effect (Prob. 6.15)
2. (20) Coulomb gauge and causality (Prob. 6.20)
3. (20) Moving electric dipole and E-field (Prob. 6.21)
4. (20) Moving electric dipole and B-field (Prob. 6.22)
5. (20) Verify the symmetry transformation properties various physical quantities in Table 7.1 of Jackson

6. (50 points) **Stokes parameters and Poincare sphere**

Consider harmonic electromagnetic plane wave

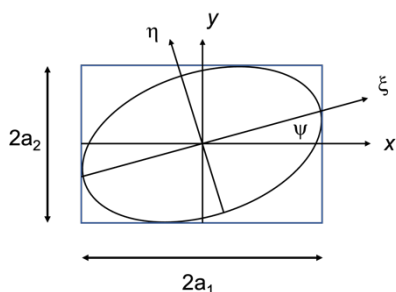
$$E_x = a_1 \cos(\vec{k} \cdot \vec{x} - \omega t + \delta_1)$$

$$E_y = a_2 \cos(\vec{k} \cdot \vec{x} - \omega t + \delta_2)$$

$$E_z = 0$$

- (a) (5) Show that the x and y components of the electric fields satisfies the following elliptic equation.

$$\left(\frac{E_x}{a_1}\right)^2 + \left(\frac{E_y}{a_2}\right)^2 - 2\left(\frac{E_x}{a_1}\right)\left(\frac{E_y}{a_2}\right)\cos(\delta_2 - \delta_1) = \sin^2(\delta_2 - \delta_1)$$



ellipse touches the sides at

$$(\pm a_1, \pm a_2 \cos(\delta_2 - \delta_1)), (\pm a_1 \cos(\delta_2 - \delta_1), \pm a_2).$$

By assigning $2a$ and $2b$ as the lengths of axes of the ellipse ($a \geq b$), one can express E_ξ and E_η as following.

$$E_{\xi} = a \cos(\vec{k} \cdot \vec{x} - \omega t + \delta_2 - \delta_1) = a \cos(\tau + \Delta), \text{ with } \Delta = \delta_2 - \delta_1,$$

and $E_{\eta} = \pm b \sin(\tau + \Delta)$.

(b) (5) Express the following four quantities in terms of $a_1, a_2, \delta_1, \delta_2, \psi$

$$a \cos(\Delta), a \sin(\Delta), \pm b \cos(\Delta), \pm b \sin(\Delta), \pm ab, \pm b/a$$

(c) (5) With the relation above, show that $a^2 + b^2 = a_1^2 + a_2^2$

(d) (5) Show the following relation holds for the ψ : $(a_1^2 - a_2^2) \sin 2\psi =$

$$2a_1 a_2 \cos \Delta \cos 2\psi$$

(e) (5) By introducing $\frac{a_2}{a_1} = \tan \alpha \left(0 \leq \alpha \leq \frac{\pi}{2}\right)$, show that $\tan 2\psi = \tan 2\alpha \cos \Delta$.

(f) (5) By introducing $\pm \frac{a_2}{a_1} = -\tan \chi \left(-\frac{\pi}{4} \leq \chi \leq \frac{\pi}{4}\right)$, show that $\sin 2\chi = \sin 2\alpha \sin \Delta$.

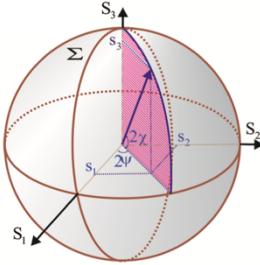
(g) (15) Using the relations derived so far, show that the Stokes parameters can be expressed in terms of χ, ψ as

$$s_0 = a_1^2 + a_2^2 = s_1^2 + s_2^2 + s_3^2$$

$$s_1 = a_1^2 - a_2^2 = s_0 \cos 2\chi \cos 2\psi$$

$$s_2 = 2 a_1 a_2 \cos \Delta = s_0 \cos 2\chi \sin 2\psi$$

$$s_3 = 2 a_1 a_2 \sin \Delta = s_0 \sin 2\chi$$



(Poincare sphere) Note that the polarization state (with the Stokes parameters) can be completely described by a point in the sphere of radius (s_0) with the angle ($2\psi, 2\chi$) with the three-perpendicular axis of s_1, s_2 & s_3 .

(f) (5) Describe the polarization states at equator, upper and lower hemisphere including north and south poles.